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Editor-in-Chief
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Dear Editor,

We submit our manuscript entitled “**Spatiotemporal spectral derivatives with forward stepwise BIC for robust partial differential equation discovery**” by Fuchang Wang and Huirong Cao for consideration as a regular article in *Applied Mathematics and Computation*.

Problem. Data-driven PDE discovery from noisy data faces two practical difficulties. First, high-order spatial derivatives such as u_{xxxx} amplify measurement noise. Second, sparse regression methods like STLSQ require a manually chosen threshold, which varies with noise level and problem. Existing spectral derivative methods are spectrally accurate but assume periodic boundaries; on non-periodic data they suffer Gibbs ringing.

What we do. We propose three connected changes:

1. **Spatiotemporal spectral derivatives.** Instead of transforming space at each time step independently, we apply a single 2D Fourier transform over the joint space-time grid. For a traveling wave $u(x, t) = f(x - ct)$, the spectrum lies along the line $k_t = -ck_x$ (Proposition 2), whereas boundary Gibbs noise spreads isotropically. A 2D filter can therefore separate signal from boundary artifacts. This is the key difference from spatial-only spectral methods (Schaeffer and McCalla, 2017).
2. **Spectral weak form.** The regression is set up in weak form: both sides of the PDE are integrated against compactly supported Gaussian test functions. This integration averages out high-frequency noise in the spectrally computed derivatives, particularly for high-order terms. We give a spectral convergence error bound (Proposition 1).
3. **Normalized forward stepwise BIC.** Each candidate term is scaled to unit ℓ_2 norm before regression. Forward stepwise selection with BIC then chooses the support automatically, with no threshold grid search. Under standard restricted-eigenvalue and minimum-coefficient conditions, the selected support is consistent (Proposition 3).

What we find. On the MDBench benchmark (four 1D PDEs at five noise levels, plus three 2D/3D problems):

- On the non-periodic advection equation—where spatial-only spectral derivatives fail completely (0/1 terms recovered)—our method recovers both terms exactly at clean data.
- On Burgers, all true terms are recovered at every noise level including SNR 10.
- On KdV and Kuramoto–Sivashinsky, exact recovery is achieved at three and two noise

levels, respectively.

- On periodic problems, accuracy is comparable to Weak SINDy while runtime is 10–20 times lower.

We compare against three baselines: finite-difference SR3, spatial-only spectral derivatives with exponential filtering, and Weak SINDy. The control-variable comparison with spatial-only spectral methods isolates the effect of the spatiotemporal transform.

Why AMC. The paper combines a numerical algorithm (spatiotemporal spectral differentiation), error analysis (three propositions), and benchmark experiments on a public dataset. It does not require GPU acceleration and runs on a standard CPU. We believe this fits the scope of *Applied Mathematics and Computation*.

This manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the submission. We declare no competing interests. Code is available at <https://github.com/fzmath/spectral-pde-discovery>.

Thank you for your consideration.

Sincerely,

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